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(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AUTOMATED WINDOW CONTROL BASED ON ENVIRONMENTAL AND SECURITY PARAMETERS A computer-implemented system for automated window control based on environmental and security parameters is disclosed. The system comprises a memory and a processor coupled to the memory, wherein the processor is configured to execute programmed instructions for monitoring ambient conditions comprising temperature data and humidity data, detecting precipitation events using moisture detection sensor data, identifying presence of objects using infrared radiation pattern analysis, sensing physical disturbances using vibrational frequency analysis, determining control actions based on sensor inputs with prioritized decision logic, and executing control actions through motor control signals to modify window position states. The system implements priority-based algorithms wherein precipitation detection overrides environmental conditions and security events prevent unauthorized access, achieving sub-second response latency for building protection. The processor-based architecture integrates environmental sensing, weather monitoring, motion detection, and vibration analysis modules with centralized decision logic providing comprehensive automated window control functionality for building automation applications. [To be published with Figure 1]
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